RDCC Summary Paper V2.0: The RDCC Inference Framework: Geometry, Topology, Likelihoods, and Euclid-Level Forecasts

Michael Lehmann[1]

[1]Independent Researcher, Relaxation-Driven Cyclic Cosmology (RDCC) Framework,
mi.lehmann@gmx.de

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Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) modifies the gravitational potential through a scale- and redshift-dependent relaxation kernel $\mathcal{R}(k, z; \alpha)$. These modifications induce geometric, morphological, and topological signatures in weak-lensing observables that require an inference framework capable of capturing structure beyond two-point statistics. Over the RDCC Companion Series (LXXX–C), we developed a complete inference ecosystem combining geometric response theory, persistent homology, transport-based topology, Minkowski functionals, emulator pipelines, multi-probe likelihoods, Fisher geometry, and degeneracy analysis.

This Summary Paper synthesizes these developments into a unified, survey-ready framework. We present: (i) the geometric and topological structure of RDCC signatures, (ii) the construction of transport-based and morphological summary statistics, (iii) the architecture of the RDCC likelihood, (iv) the Fisher geometry and information-flow structure underlying RDCC inference, (v) optimal RDCC probes and survey strategies, and (vi) Euclid-level forecasts for the relaxation parameter α. This document serves as the central reference for the RDCC inference ecosystem and provides a coherent foundation for future observational applications.

1 Introduction

The Relaxation-Driven Cyclic Cosmology (RDCC) introduces a dynamical relaxation mechanism that modifies the gravitational potential through a single parameter α. These modifications produce scale-dependent distortions in the lensing potential that propagate into geometric, morphological, and topological features of the weak-lensing convergence field. Capturing these signatures requires an inference framework that extends beyond two-point statistics and incorporates non-Gaussian structure, geometric sensitivity, and topological information.

Over the RDCC Companion Series (LXXX–C), we developed such a framework. It combines geometric response fields, persistent homology, transport-based topology, Minkowski functionals, emulator pipelines, multi-probe likelihoods, Fisher geometry, and degeneracy analysis into a coherent and modular ecosystem.

The RDCC inference framework rests on four conceptual pillars:

1. **Geometric Signatures:** RDCC induces scale-dependent distortions in the lensing potential, encoded in geometric response fields, Fisher channels, and curvature-aware parameter-space metrics.

2. **Topological Structure:** RDCC modifies the birth–death structure of persistent homology, the morphology of peaks and voids, and the transport geometry of topological summaries.
3. **Likelihood Architecture:** A modular likelihood integrates transport-based statistics, Minkowski functionals, peak/void statistics, power spectra, and neural-compressed summaries, supported by emulator pipelines and simulation-based calibration.
4. **Forecasts and Model Selection:** Euclid-level forecasts, Fisher-channel analysis, and evidence decomposition quantify the detectability of RDCC and its distinguishability from ΛCDM.

This Summary Paper synthesizes these components into a unified narrative. Rather than reproducing the detailed derivations contained in the individual Companions, we highlight the structural logic of the RDCC inference framework, the interplay between geometry and topology, and the methodological innovations that enable robust and high-fidelity parameter estimation.

2 The RDCC Model and Its Geometric Structure

RDCC modifies the gravitational potential through a relaxation kernel

$$\Phi_{\mathrm{RDCC}}(k,z) = \Phi_{\Lambda\mathrm{CDM}}(k,z)\,\mathcal{R}(k,z;\alpha), \tag{1}$$

where $\mathcal{R}(k,z;\alpha)$ introduces scale-dependent deviations that become relevant at $k \gtrsim 0.1\,h\,\mathrm{Mpc}^{-1}$. These deviations propagate into weak-lensing observables and generate geometric and topological signatures that form the basis of RDCC inference.

2.1 Geometric Response Fields

For an observable O, the geometric response to α is defined as

$$\mathcal{R}_O = \frac{\partial O}{\partial \alpha}, \tag{2}$$

which quantifies the local deformation induced by RDCC. These response fields capture peak sharpening, void broadening, filament-void boundary shifts, and curvature changes in the lensing potential.

2.2 Fisher Geometry

The RDCC parameter space inherits a Riemannian structure from the Fisher information metric

$$g_{\alpha\alpha} = \mathbb{E}\left[\left(\frac{\partial \log \mathcal{L}}{\partial \alpha}\right)^2\right], \tag{3}$$

which encodes the curvature of the likelihood surface. Fisher geometry reveals RDCC-sensitive directions, degeneracy manifolds, and natural coordinates for efficient inference.

3 Topological and Transport-Based RDCC Signatures

Weak-lensing observables contain rich geometric and topological information that extends beyond two-point statistics. RDCC induces characteristic distortions in the morphology of the convergence field, the connectivity of the cosmic web, and the birth–death structure of persistent homology. These effects motivate the use of topological and transport-based summary statistics, which provide noise-resistant and non-Gaussian probes of RDCC signatures.

Companion	Content	Role in RDCC Framework
LXXX–LXXXII	Geometric RDCC signatures, response fields, scale-dependent distortions	Defines geometric foundations; identifies RDCC-sensitive scales and response patterns
LXXXIII–LXXXVI	Topological statistics, persistent homology, transport geometry	Establishes non-Gaussian RDCC signatures; provides robust topology-based probes
LXXXVII–LXXXVIII	Minkowski functionals, morphological descriptors	Captures RDCC-induced shape and connectivity changes in convergence maps
LXXXIX–XC	Emulator pipelines, simulation grids, interpolation architecture	Enables fast likelihood evaluation; propagates simulation uncertainty
XCI–XCII	Forecasts, Euclid-like constraints, multi-probe combinations	Quantifies detectability of α; identifies optimal probe combinations
XCIII–XCIV	Fisher geometry, curvature, natural coordinates	Defines RDCC parameter-space geometry; identifies sensitivity directions
XCV–XCVI	Information tensor, Fisher channels, information flow	Decomposes RDCC information; identifies dominant channels and null directions
XCVII–XCVIII	Degeneracy structure, null space, RDCC–ΛCDM alignment	Maps degeneracy manifolds; identifies blind modes and orthogonal probes
XCIX	Optimal probes, survey strategies, efficiency function	Ranks probes; provides survey recommendations for RDCC detection
C	Benchmark Suite, validation, stress tests, reproducibility framework	Ensures robustness, emulator validation, and cross-probe consistency

Table 1: Roadmap of the RDCC Companion Series. Each Companion contributes a specific conceptual or technical component to the RDCC inference ecosystem.

3.1 Persistent Homology and Birth–Death Structure

Persistent homology characterizes the topology of the convergence field by tracking the evolution of connected components, loops, and voids across thresholds. For a filtration parameter ν, the persistence diagram $\mathcal{D}(\nu)$ encodes the birth and death of topological features.

RDCC modifies the persistence structure through:

- enhanced persistence of filamentary loops,
- earlier birth of small-scale peaks,
- delayed death of large voids,
- scale-dependent shifts in birth–death trajectories.

These effects are captured by the RDCC topological gradient

$$\mathcal{G}_{\text{topo}} = \frac{\partial \mathcal{T}}{\partial \alpha}, \tag{4}$$

where $\mathcal{T}$ denotes a transport-based topological statistic.

3.2 Transport Geometry of Topological Statistics

Transport-based statistics quantify the geometric displacement between persistence diagrams. Let $\mathcal{D}_{\mathrm{RDCC}}$ and $\mathcal{D}_{\Lambda\mathrm{CDM}}$ denote diagrams under RDCC and ΛCDM. Their quadratic optimal-transport distance is

$$W_2(\mathcal{D}_{\mathrm{RDCC}}, \mathcal{D}_{\Lambda\mathrm{CDM}})\,, \tag{5}$$

which measures the minimal geometric deformation required to transform one diagram into the other.

RDCC induces coherent transport flows that reflect:

- peak migration toward higher persistence,
- void expansion and boundary smoothing,
- filament thickening or thinning,
- redistribution of topological mass across scales.

These flows provide a geometric representation of RDCC-induced topological deformations.

3.3 Minkowski Functionals and Morphological Signatures

Minkowski functionals $V_i(\nu)$ capture the morphology of the convergence field across thresholds. RDCC modifies:

- the area functional V_0 through small-scale suppression,
- the boundary length V_1 via filament–void transitions,
- the Euler characteristic V_2 through changes in peak/void counts.

These morphological signatures complement persistent homology and provide summary statistics that are computationally efficient and robust to noise.

3.4 Joint Topological–Geometric Structure

Topological and geometric signatures are not independent. RDCC induces coherent patterns across:

- geometric response fields,
- transport flows in persistence space,
- Minkowski functional derivatives,
- Fisher-channel alignments.

This joint structure is essential for constructing multi-probe likelihoods and for identifying optimal RDCC probes.

3.5 Summary

Topological and transport-based statistics provide a powerful and noise-resistant characterization of RDCC signatures. They capture the non-Gaussian structure of the convergence field, reveal scale-dependent deformations induced by the relaxation mechanism, and form a central component of the RDCC inference framework. Their integration with geometric sensitivity maps and Fisher geometry enables a comprehensive and robust analysis of RDCC effects.

4 The RDCC Likelihood Architecture

The RDCC inference framework requires a likelihood capable of integrating non-Gaussian summary statistics, topological descriptors, geometric sensitivity patterns, and emulator-based predictions. The resulting architecture is modular, hierarchical, and designed to combine heterogeneous probes while maintaining statistical consistency and computational efficiency.

4.1 Multi-Probe Structure

Let $\mathcal{S}$ denote the full set of RDCC summary statistics, including:

- power spectra and correlation functions,
- Minkowski functionals,
- transport-based topological statistics,
- peak and void counts,
- neural-compressed summaries.

The joint likelihood factorizes as

$$\mathcal{L}(\mathcal{S} \mid \alpha, \theta) = \prod_i \mathcal{L}_i(S_i \mid \alpha, \theta), \tag{6}$$

where θ denotes the ΛCDM parameters. Each component likelihood $\mathcal{L}_i$ is calibrated using simulation-based inference and validated through cross-probe consistency tests.

4.2 Emulator-Based Predictions

High-fidelity RDCC simulations are computationally expensive. To enable efficient likelihood evaluation, we employ emulator pipelines trained on simulation grids and validated using the RDCC Benchmark Suite (Companion C).

For each summary statistic S_i, the emulator predicts

$$S_i(\alpha, \theta), \tag{7}$$

with uncertainty propagated into the likelihood via

$$\Sigma_i = \Sigma_{\rm sim} + \Sigma_{\rm emul}. \tag{8}$$

This ensures that emulator uncertainty does not bias inference or artificially inflate evidence contrast.

4.3 Transport-Based Likelihood Components

Transport-based statistics require likelihoods that respect the geometry of persistence diagrams. For a quadratic optimal-transport distance W_2, we define

$$\mathcal{L}_{\rm topo} \propto \exp\left[-\frac{W_2(\mathcal{D}_{\rm data}, \mathcal{D}(\alpha))}{2\,\sigma^2_{W_2}}\right], \tag{9}$$

which captures geometric deformations in topological space and is robust to noise and sampling variance.

4.4 Neural Compression and Likelihood-Free Components

For high-dimensional statistics, we employ neural compression to construct compressed summaries z that retain maximal information about α:

$$z = f_{\mathrm{enc}}(\mathcal{S}), \tag{10}$$

where f_{enc} is trained to maximize Fisher information or minimize likelihood-free loss functions.

The compressed likelihood is then

$$\mathcal{L}(z \mid \alpha) = \mathcal{N}(z;\, \mu(\alpha),\, \Sigma_z)\,, \tag{11}$$

with Σ_z estimated via simulation-based calibration.

4.5 Cross-Probe Covariance Structure

The full covariance matrix is

$$C = C_{\mathrm{stat}} + C_{\mathrm{sys}} + C_{\mathrm{cross}}, \tag{12}$$

where C_{cross} captures correlations between geometric, topological, and morphological probes.

RDCC induces characteristic cross-covariance patterns, including:

- alignment between geometric response fields and topological flows,
- covariance suppression in degeneracy-aligned directions,
- enhanced cross-probe coherence at RDCC-sensitive scales.

4.6 Summary

The RDCC likelihood architecture integrates geometric, topological, and compressed statistics into a unified multi-probe framework. Emulator-based predictions, transport-aware likelihoods, and cross-probe covariance modeling ensure robustness and flexibility. This architecture forms the computational core of RDCC inference and underpins the forecasts and model-selection results presented in later sections.

5 Fisher Geometry, Information Flow, and Degeneracy Structure

The RDCC inference framework is governed by a geometric structure induced by the Fisher information metric, which encodes the curvature of the likelihood surface and determines the efficiency of parameter estimation. Combined with observable gradients and topological response fields, the Fisher geometry defines the information flow from weak-lensing observables to RDCC parameter constraints.

5.1 Fisher Information and the RDCC Metric

For a parameter vector (α, θ), the Fisher information matrix is

$$G_{ij} = \mathbb{E}\left[\frac{\partial \log \mathcal{L}}{\partial \theta_i}\frac{\partial \log \mathcal{L}}{\partial \theta_j}\right], \tag{13}$$

where θ denotes the ΛCDM parameters. The Fisher metric defines a Riemannian geometry on parameter space, with geodesics corresponding to directions of minimal information loss.

For RDCC, the Fisher metric reveals:

- strong curvature in RDCC-sensitive directions,
- flat regions corresponding to degeneracy manifolds,
- natural coordinates that diagonalize information channels,
- geometric pathways linking RDCC and ΛCDM variations.

These geometric features guide the construction of efficient samplers and inform the design of optimal probes.

5.2 Observable Gradients and Information Flow

Let S_i denote a summary statistic. Its RDCC gradient is

$$\mathcal{G}_i = \frac{\partial S_i}{\partial \alpha}. \tag{14}$$

The information tensor introduced in Companion XCVII is

$$I_{ij} = \mathbb{E}[\mathcal{G}_i \, \mathcal{G}_j] \,, \tag{15}$$

which generalizes the Fisher metric to the space of summary statistics.

Diagonalizing I yields Fisher channels:

$$I \, v_k = \lambda_k \, v_k. \tag{16}$$

Each channel v_k represents:

- a direction of coherent RDCC response,
- a geometric mode of information flow,
- a contribution λ_k to the total Fisher information,
- a pathway linking observables to evidence.

Large eigenvalues correspond to dominant RDCC information pathways, while small eigenvalues indicate suppressed or degenerate directions.

5.3 Degeneracy Structure and Null Directions

Not all directions in summary-statistic space carry RDCC information. The RDCC Null Space is defined as

$$\mathcal{N} = \left\{ v \,\middle|\, v^T I v = 0 \right\}. \tag{17}$$

Vectors in $\mathcal{N}$ correspond to:

- blind modes with zero RDCC sensitivity,
- combinations of observables aligned with ΛCDM variations,
- directions dominated by noise or cosmic variance,
- degeneracy manifolds in Fisher geometry.

These null directions play a central role in probe optimization and survey design, as they identify observables that do not contribute to RDCC inference.

5.4 Geometric Interpretation of Degeneracy Manifolds

In Fisher geometry, degeneracy manifolds satisfy

$$g_{\alpha\alpha}\, d\alpha^2 = g_{\theta\theta}\, d\theta^2, \tag{18}$$

indicating that variations in α can be mimicked by variations in ΛCDM parameters.

These manifolds represent:

- curved surfaces of equal RDCC sensitivity,
- geodesic alignments between RDCC and ΛCDM,
- regions of suppressed evidence contrast,
- pathways of minimal distinguishability.

Understanding these manifolds is essential for breaking degeneracies and designing optimal RDCC probes.

5.5 Summary

Fisher geometry, information flow, and degeneracy structure form the mathematical backbone of RDCC inference. Observable gradients propagate through the information tensor to define Fisher channels, while degeneracy manifolds identify directions of suppressed sensitivity. Together, these structures guide the construction of efficient likelihoods, optimal probes, and robust forecasting pipelines.

6 Optimal Probes and Survey Strategies

The RDCC inference framework integrates geometric sensitivity, topological structure, and Fisher-channel analysis to identify the most informative observables for constraining the relaxation parameter α. This section summarizes the optimal RDCC probes and the survey strategies that maximize detectability and minimize degeneracy with ΛCDM.

6.1 The RDCC Probe Efficiency Function

To quantify the information content of a summary statistic S, we define the RDCC Probe Efficiency Function

$$\varepsilon(S) = (1 - D(S))\,\frac{\mathcal{I}(S)}{\mathrm{Var}(S)}, \tag{19}$$

where $D(S)$ is the degeneracy factor derived from the RDCC Null Space (Companion XCVIII), and $\mathcal{I}(S)$ denotes the Fisher information contributed by S.

High values of $\varepsilon(S)$ indicate probes that:

- respond strongly to RDCC-induced distortions,
- break degeneracies with ΛCDM parameters,
- align with dominant Fisher channels,
- exhibit stable topological or geometric signatures.

This efficiency function provides a unified ranking of RDCC probes.

6.2 Ranking RDCC Probes

Evaluating $\varepsilon(S)$ across geometric, topological, and morphological statistics yields a consistent hierarchy:

- **Transport-based topological statistics** provide the strongest RDCC sensitivity, particularly at intermediate persistence scales.
- **Minkowski functionals** capture morphological distortions induced by RDCC and complement topological probes.
- **Peak and void statistics** are highly sensitive to relaxation-driven shifts in the potential.
- **Power spectra** contribute primarily at large scales but are partially degenerate with ΛCDM variations.
- **Neural-compressed summaries** efficiently combine multiple probes and retain high Fisher information.

This ranking reflects the multi-scale and non-Gaussian nature of RDCC signatures.

6.3 Fisher-Channel Alignment

The geometric alignment of a probe S with the dominant Fisher channel $v_{\max}$ is

$$A(S) = \frac{\left\langle \frac{\partial S}{\partial \alpha}, v_{\max} \right\rangle}{\left\| \frac{\partial S}{\partial \alpha} \right\| \, \| v_{\max} \|}. \tag{20}$$

Probes with high alignment:

- efficiently trace the dominant RDCC information pathway,
- minimize curvature-induced sampling inefficiencies,
- reduce the impact of degeneracy manifolds,
- provide stable likelihood gradients.

Transport-based topological statistics exhibit the highest alignment, followed by Minkowski functionals and peak statistics.

6.4 Breaking Degeneracies with ΛCDM

Degeneracy manifolds identified in Companion XCVIII reveal directions in summary-statistic space where RDCC variations mimic ΛCDM parameter shifts. Optimal probes are those that:

- have minimal projection onto the RDCC Null Space,
- exhibit strong orthogonality to ΛCDM-aligned subspaces,
- amplify RDCC-specific geometric or topological distortions.

Topological probes are particularly effective at breaking degeneracies due to their sensitivity to non-Gaussian structure.

6.5 Survey Strategies for Maximizing RDCC Detectability

The RDCC probe hierarchy informs survey strategies that maximize sensitivity to α. Key recommendations include:

- **Angular resolution:** prioritize intermediate scales where RDCC-induced topological distortions are strongest.
- **Tomographic binning:** allocate finer redshift resolution to bins with high Fisher-channel alignment.
- **Noise mitigation:** emphasize probes with stable topological persistence under shape noise.
- **Cross-probe integration:** combine geometric and topological probes to suppress degeneracy directions.
- **Emulator fidelity:** ensure high accuracy for probes with large $\varepsilon(S)$ to avoid information loss.

These strategies are compatible with Euclid-like survey specifications and provide a roadmap for future RDCC analyses.

6.6 Summary

Optimal RDCC probes are those that combine strong geometric or topological sensitivity with minimal degeneracy and high Fisher-channel alignment. The resulting probe hierarchy and survey strategies maximize RDCC detectability and provide a principled foundation for high-precision cosmological inference.

7 Forecasts and Model Selection

The RDCC inference framework enables quantitative forecasts for the relaxation parameter α and provides a principled basis for model comparison with ΛCDM. These forecasts combine geometric sensitivity, topological statistics, emulator-based predictions, and the multi-probe likelihood architecture described in previous sections.

7.1 Euclid-Level Constraints on α

Using the full RDCC likelihood, including transport-based topology, Minkowski functionals, peak statistics, and neural-compressed summaries, we obtain forecasted constraints of the form

$$\sigma(\alpha) \sim \mathcal{O}(10^{-2}), \tag{21}$$

with the precise value depending on the probe combination and tomographic binning.

The strongest constraints arise from:

- intermediate-scale transport-based topological statistics,
- peak and void counts sensitive to RDCC-induced potential shifts,
- neural-compressed summaries that combine geometric and topological information.

Power spectra alone provide significantly weaker constraints due to partial degeneracy with ΛCDM parameters.

7.2 Fisher-Channel Contributions to Forecasts

The Fisher-channel decomposition (Companion XCVII) reveals that RDCC information is concentrated in a small number of dominant channels. The leading channel typically accounts for

$$\sim 70\% \tag{22}$$

of the total Fisher information, with the remaining channels capturing subdominant geometric and topological features.

This structure explains:

- the efficiency of neural compression,
- the robustness of topological probes,
- the limited contribution of high-dimensional statistics,
- the sensitivity of forecasts to degeneracy directions.

7.3 Degeneracy Structure and Forecast Stability

Degeneracy manifolds identified in Companion XCVIII reveal directions in parameter space where RDCC variations mimic ΛCDM shifts. These degeneracies affect forecast stability by:

- reducing sensitivity in null directions,
- suppressing evidence contrast in aligned subspaces,
- amplifying the importance of topological probes,
- increasing the role of cross-probe covariance.

Forecasts that rely solely on geometric probes are more susceptible to these degeneracies, while topological probes provide strong orthogonality to ΛCDM-aligned directions.

7.4 Evidence Decomposition and Model Selection

The RDCC evidence can be decomposed into Fisher-channel contributions:

$$\Delta \log Z_k = \frac{1}{2} \lambda_k, \tag{23}$$

where λ_k is the eigenvalue of the k-th Fisher channel. Summing over channels yields the total evidence contrast between RDCC and ΛCDM.

Topological probes contribute disproportionately to $\Delta \log Z$, reflecting their sensitivity to non-Gaussian RDCC signatures. Geometric probes contribute primarily through the leading Fisher channel, while power spectra contribute weakly due to degeneracy with ΛCDM.

7.5 Combined Forecasts and Model-Selection Prospects

Combining all probes yields:

- tight constraints on α,
- strong evidence contrast for moderate RDCC amplitudes,
- robustness to noise and systematics,
- stability under cross-probe consistency tests.

The RDCC inference framework therefore provides a powerful and flexible approach to testing relaxation-driven cosmology with upcoming weak-lensing surveys.

7.6 Summary

Forecasts and model-selection analyses demonstrate that RDCC induces detectable geometric and topological signatures in weak-lensing observables. Topological probes, Fisher-channel structure, and degeneracy analysis play central roles in achieving high sensitivity and robust model discrimination. These results establish RDCC as a testable extension of ΛCDM and motivate future observational applications.

8 Conclusion

The Relaxation-Driven Cyclic Cosmology (RDCC) introduces a scale-dependent relaxation mechanism that produces geometric and topological signatures in weak-lensing observables. Over the course of the RDCC Companion Series (LXXX–C), we developed a complete inference framework capable of capturing these signatures across multiple statistical domains. This Summary Paper synthesized the conceptual foundations, methodological components, and forecasting results of that framework.

RDCC induces coherent deformations in the gravitational potential, which propagate into geometric response fields, Fisher-channel structure, and persistent topological features. These signatures motivate the use of transport-based topology, Minkowski functionals, peak and void statistics, and neural-compressed summaries. The resulting multi-probe likelihood architecture integrates these heterogeneous observables into a unified inference pipeline, supported by emulator-based predictions and validated through the RDCC Benchmark Suite.

Fisher geometry and information-flow analysis reveal that RDCC information is concentrated in a small number of dominant channels, while degeneracy manifolds identify directions of suppressed sensitivity and partial alignment with ΛCDM variations. Optimal probes are those that combine strong geometric or topological response with minimal degeneracy and high Fisher-channel alignment. These probes form the basis of Euclid-level forecasts, which demonstrate that RDCC induces detectable signatures and that the relaxation parameter α can be constrained with high precision.

The RDCC inference framework is modular, extensible, and designed for application to upcoming weak-lensing surveys. Its combination of geometric analysis, topological structure, and simulation-based inference provides a robust foundation for testing relaxation-driven cosmology. Future work will extend this framework to cross-correlations with galaxy clustering, multi-wavelength probes, and joint analyses with dynamical tracers. The RDCC Companion Series serves as a technical backbone for these developments and ensures reproducibility, consistency, and methodological clarity across the entire RDCC ecosystem.